NASA Data in Space Mission Designer

Team Shay Rafe3 · NASA Space Apps Challenge 2026 · every NASA resource we used, where it is in the game, the science and the logic

> **Rule:** every number lives in `game/scripts/data.gd` with a `src` field. Real values name their source. Invented values say `"game rule"` or `"game estimate"`. The physics is in `game/scripts/game.gd`.

1. Official challenge resources

Resource	Used for	Where in game	Link
NASA *Design and Develop Science Missions* (2025)	4 mission segments → our 4 roles; goal-first design	Builder role tabs, `ROLES`, `DEST.objectives`	nasa.gov/.../250323-design-and-develop-science-mission.pdf
NASA GMAT	Engine Isp 300 s; solar arrays lose 5%/year	`ISP`, `SOLAR_DECAY_PER_YEAR`	software.nasa.gov/software/GSC-19468-1
JPL Concept Maturity Levels (2013)	35% margins; 3 dB link; green/yellow/red checks	`MARGIN`, builder lamps, GO/NO-GO poll	dataverse.jpl.nasa.gov/...hdl:2014/44299
Goddard Mission Design Lab	Engineering seats grouped into 4 roles; parts list = equipment list	Roles, builder parts bin	etd.gsfc.nasa.gov/.../mission-design-lab/
NASA Mission Viz	Ideas only (no licence) for the transfer-orbit map	Cruise map, `flight.gd`	nasa.github.io/mission-viz
NASA Webb *Build It Yourself*	"Your design is like mission X" ending	Report: real-mission comparison, `DEST.real`	science.nasa.gov/mission/webb/game-build-it-yourself
NASA CubeSat 101 (2017)	Failure lessons, keep reserves	Not much yet (planned hints)	nasa.gov/.../nasa_csli_cubesat_101_508.pdf
Space Apps 2023 VR mission	Mission timeline idea	Ideas only	spaceappschallenge.org/2023/...
CSA Digital Games	Looked at for ideas	Not used	asc-csa.gc.ca/.../digital-games-and-activities.asp

2. Extra NASA data in the game

Data	Real values	Where in game	Link
NASA DONKI (space weather)	15 real geomagnetic storms, 2024. Strongest Kp 9 (10 May 2024)	`donki_data.gd`, solar-storm event	api.nasa.gov (DONKI)
NASA planetary fact sheets	Gravity, sunlight W/m², distance, closest approach for 8 bodies	`DEST[*]`	nssdc.gsfc.nasa.gov/planetary/factsheet
NASA radio light delay	Mars 4–24 min	`DEST.light_delay`, report	nasa.gov/.../space-communications-7-things...
NASA MMRTG fact sheet	110 W, 45 kg, 14-year life	`PARTS.rtg`, `RTG_LIFE_YEARS`	mars.nasa.gov/.../MMRTG_FactSheet...pdf
NASA Small Spacecraft Tech 2026, ch. 3	Solar ≈30 W/kg (1–200); Li-ion 150–270 Wh/kg	`PARTS.solar*`, `battery*`	nasa.gov/smallsat-institute/sst-soa/power-subsystems
NASA NSSDCA: MRO	3 m dish, up to 6 Mbps from Mars	`PARTS.hga`	nssdc.gsfc.nasa.gov/...id=2005-029A
NASA NSSDCA: LRO	Ka-band 100–300 Mbps	`PARTS.ka`	nssdc.gsfc.nasa.gov/...id=2009-031A
MRO instruments	CTX 3 kg, 7 W · CRISM 32.9 kg, 44–47 W, 2–2.5 Gb/look	`PARTS.camera`, `spectro`	science.nasa.gov/.../mro/science-instruments

Cassini MAG	3.0 kg, 3.1 W	`PARTS.mag`	science.nasa.gov/.../magnetometer/mag-technical-write-up
JPL Juno radiation vault	172 kg titanium, cuts radiation 800×	`PARTS.shield`, Jupiter arrival	jpl.nasa.gov/news/...juno...fireworks...
Perseverance (Mars 2020)	203-day cruise; chute 1,512 → 306 km/h; 1,025 kg; MMRTG	`DEST.mars`, `PARTS.chute`	science.nasa.gov/mission/mars-2020-perseverance
Apollo 11	3-day trip; 0.89 km/s lunar braking; 21.5 kg rock; ascent stage	`DEST.moon`, `PARTS.ascent`	nasa.gov/mission/apollo-11
CAPSTONE	25 kg, $9.95M launch on Electron, 4.5-month path	`ROCKETS.electron`, Moon low-energy path	nasa.gov/news-release/...cubesat-to-moon...
Cassini	6.7-year gravity-assist path; 3 RTGs; Grand Finale ring dives	`DEST.saturn`, drone, `PARTS.dust`	science.nasa.gov/mission/cassini
MESSENGER	6.6-year path, 6 flybys; 862 m/s orbit burn; sunshade	`DEST.mercury`, `PARTS.sunshade`	science.nasa.gov/mission/messenger
Juno	4.9-year path; 542 m/s orbit burn	`DEST.jupiter`	science.nasa.gov/mission/juno
Voyager 2	Uranus 8.4 yr, Neptune 12 yr	`DEST.uranus, neptune`	science.nasa.gov/mission/voyager
Europa Clipper	≈6,000 kg on Falcon Heavy to Jupiter	`ROCKETS.heavy`	science.nasa.gov/mission/europa-clipper
Spirit & Opportunity	Airbag landings (2004)	`PARTS.airbags`	science.nasa.gov/.../spirit-and-opportunity
Sojourner · Curiosity	11.5 kg · 899 kg rovers (anchors)	`PARTS.rover`, `rover_big`	science.nasa.gov/mission/mars-pathfinder · .../msl-curiosity
Venera 13 (NSSDC)	127 min alive on Venus; 122-day trip	`DEST.venus`, `PARTS.hull`	nssdc.gsfc.nasa.gov/...id=1981-106D
NASA Discovery 2019 call	25% cost reserve	`MARGIN.budget`	astrobiology.nasa.gov/...discovery-program...
ESA (partner): Venus Express, BepiColombo	153-day trip, 1.25 km/s burn · 8.1-year path, 9 flybys	`DEST.venus, mercury`	esa.int/.../Venus_Express · .../BepiColombo

Rocket prices and lift (Electron, Falcon 9, Vulcan, Falcon Heavy) come from the makers and Wikipedia, labelled as secondary in `ROCKETS`.

3. The science and the logic

Topic	Science	Game logic	Code
Fuel	Rocket equation $\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_1)$, Isp 300 s, g_0 9.81	Fuel needed = `dry·(e^(Δv/v_e)−1)`. Δv = cruise 0.03 + orbit braking + landing	`rocket_eq_prop()`, `burn()`, `dv_plan()`
Solar power	Sunlight falls with distance² from the Sun; arrays age	`P = P₁AU / d² · 0.95^years · dust`. Too little in cruise → "OUT OF POWER"	`calc()`, `phase_cruise()`
Nuclear power	MMRTG gives 110 W at any distance, 14-year life	Constant 110 W; half power after 14 years (game rule)	`calc()`
Battery	Night or battery-only landers drain stored energy	`need = (draw − RTG) · dark hours`; Mars night 12.3 h; Venus 127 min	`calc()`
Radio link	Signal falls with distance² from Earth	`rate = dish rate / d²` (capped) × contact time (8 h/day, game rule)	`calc()`, `phase_home()`
Margins	JPL CML 4: 35% mass, power, energy; 3 dB deep-space link; Discovery: 25% cost reserve	A bar is green only with that much spare. 3 dB = room to send 2× the data	`MARGIN`, `_use()`, `role_issues()`

Rocket lift	Each launcher lifts a fixed mass to each target; faster paths lift less	Ship mass > capacity → "TOO HEAVY" at launch	`rocket_cap()`, `phase_launch()`
Paths	Gravity assists and low-energy paths save fuel but take longer	Each path has real days + fuel/lift multipliers	`DEST.traj`
Landing	Real surface gravity; no air = engine only; Mars = heat shield + chute	Live landing with real g; legs 4 m/s, airbags 12 m/s (game rules)	`landing_setup()`, `landing.gd`
Heat	Near the Sun light is much stronger (Mercury 9,083 W/m²)	Over 5,000 W/m² without sunshade → "OVERHEATED"	`needs_shade()`
Radiation	Jupiter's belts kill electronics (Juno's vault)	No shield → comms or science part breaks	`phase_arrive()`
Solar storms	Kp index; Kp ≥ 7.5 = severe storm	A random real DONKI storm. Severe: 70% damage chance without shield; else 15%	`_dress_storm()`
Time	Mission time = development + cruise + operations	200 days + 30 per part kind + 300 for nuclear approval (game rules)	`calc()`

4. Destinations (NASA fact sheets + real missions)

Place	Gravity m/s²	Sunlight W/m²	Sun AU	Orbit braking km/s	Paths (real anchor)
Moon	1.62	1,361	1.0	0.90 (Apollo 11)	3 d Apollo · 138 d CAPSTONE
Mercury	3.70	9,083	0.387	0.86 (MESSENGER)	6.6 yr MESSENGER · 8.1 yr BepiColombo
Venus	8.87	2,601	0.723	1.25 (Venus Express)	153 d Venus Express · 122 d Venera 13
Mars	3.73	586	1.52	2.1 (Hohmann maths)	203 d Perseverance · 150 d fast
Jupiter	25.92	50.3	5.20	0.54 (Juno)	4.9 yr Juno · 2.7 yr Hohmann
Saturn	11.19	14.8	9.54	0.63	6.7 yr Cassini · 6.1 yr Hohmann
Uranus	9.01	3.69	19.19	1.09 (Uranus Orbiter study)	13 yr study · 8.4 yr Voyager 2
Neptune	11.27	1.51	30.07	2.5 (Ice Giants study, est.)	12 yr Voyager 2 · 16 yr Odyssey study

5. Real mission stories behind the event cards

37 event cards in `game/scripts/events.gd`; each shows "this really happened" with its source. Chance and damage depend on the player's design.

- **Cruise:** Mars Odyssey MARIE storm (2003) · Webb micrometeoroid (2022) · Galileo antenna (1991) · Mars Climate Orbiter units (1999) · Voyager 2 receiver (1978) · MESSENGER sunshade · Dawn & Kepler wheels · CAPSTONE tumble (2022) · Akatsuki engine (2010) · BepiColombo thrusters (2024)
- **At the site:** Odysseus tipped (2024) · Curiosity wheels, backup computer (2013) · Spirit sand trap (2009) · Opportunity dust storm (2018) · InSight panel cleaning · Perseverance empty tube (2021) · Venera 13/14 · Cassini ring dives (2017) · Juno valves (2016), camera (2023) · Galileo recorder (1995) · Voyager 2 platform (1981) · Huygens relay (2000)
- **Going home:** Apollo 13 tank (1970) · Genesis switches (2004) · Hayabusa leak (2005) · Voyager 2 at Uranus and Neptune · Hubble one-gyro (2024) · New Horizons safe mode (2015) · Voyager 1 chip (2023) · MESSENGER helium (2015)

6. What we invented (labelled "game rule" in code)

Part prices, bus mass, antenna base rates, 8 h/day of contact, legs 4 m/s, airbags 12 m/s, development days, budgets and time limits, difficulty settings, the drone, scoring and grades, multiplayer timers and weights. Everything else comes from the sources above.